

Question and Answers

Apache Logs

2. How many lines in apache_access.log have status 404?

Answer: 8

3. List every IP address that caused a 404 — and the path it tried to access

Answer:
203.0.113.9 → /admin, /admin/config.php, /.env, /wp-admin/, /phpmyadmin/ — 5 hits (Nikto scanner)
198.51.100.77 → /sitemap.xml, /.git/config, /backup.zip — 3 hits (masscan)

4. Which IP caused the most 404 errors?

Answer: Most 404s from: 203.0.113.9

5. What is the IP, path, and byte count of the largest response?

Answer Format: IP | Path | bytes

Answer: 192.168.1.45 | /images/banner.jpg | 94,125 bytes

6. Why might a very large response be interesting from a security perspective?

Answer:
1. Data Exfiltration: An attacker successfully downloading a large database or sensitive file.
2. Successful Reconnaissance: The discovery of a large backup or resource that shouldn't be public.
3. Denial of Service (DoS): Requests for resource-intensive files, like /slow-query (90,277 bytes), that can strain server memory or bandwidth.

7. Which User-Agent string is clearly an automated scanner?

Answer:
Nikto/2.1.6
masscan/1.3.2

8. From which IP did that User-Agent appear?

Answer:
Nikto/2.1.6 from 203.0.113.9 — web vulnerability scanner
masscan/1.3.2 from 198.51.100.77 — port/path scanner

9. How can you tell it is a scanner and not a real browser?

Answer:
1. Automated User-Agents: Scanners often identify as tools rather than browsers, such as masscan/1.3.2 or Nikto/2.1.6.
2. Probing Patterns: Scanners request common sensitive files that a normal user wouldn't look for, such as /.env, /.git/config, or /wp-admin/.
3. High Error Rate: Scanners often trigger multiple 404 Not Found errors in a very short time as they "guess" file paths.

DNS Logs

10. Which client IP uses source port 9999

Answer: 198.51.100.77

11. List every NXDOMAIN response from dns.log — domain name and client IP

Answer:
203.0.113.9 queried 4 internal names (internal-api.corp.local, fileserver, backup, devops)
198.51.100.77 queried c2-server-xyz123.ru and ab7f3kqp.top

12. Do any of the queried domains look suspicious? Why?

Answer:

203.0.113.9 queried 4 internal names (internal-api.corp.local, fileserver, backup, devops) – all NXDOMAIN. Classic internal recon pattern.
198.51.100.77 queried c2-server-xyz123.ru and ab7f3kqp.top – both NXDOMAIN. Looks like a DGA/C2 callback failure.

SSH Logs

13. List every IP with at least one failed SSH login and its failure count

Answer: 203.0.113.9 → 9 failures targeting users: root, admin, ubuntu, pi

14. What happened to the attacker after the maximum login attempts were exceeded?

Answer: Gets disconnected with "Too many authentication failures"

15. List every successful SSH login: for each line with Accepted, give the username, source IP

Answer: miko / 192.168.1.45; admin / 10.0.0.23; svc_backup / 10.0.0.50; deploy / 10.0.0.23.

SMTP Logs

16. Which IP addresses were rejected in mail.log?

Answer:

203.0.113.9 rejected twice (different recipients: root@corp.local, postmaster@corp.local)
198.51.100.77 rejected once (info@corp.local)

17. What does '554 5.7.1 Relay access denied' mean in plain English?

Answer:

1. Unauthorized Sender: The server does not recognize the sender's IP address or credentials as part of its internal network or allowed list.
2. External Recipient: The sender is trying to use the server as a "middleman" to send mail to a destination the server isn't responsible for.
3. Anti-Spam Measure: This is a security feature to prevent "Open Relaying," which stops spammers from using the server to send unsolicited bulk emails.

18. Find the message that ends with status=bounced. What recipient address was used?

Answer: nobody@nonexistent.tld

Cross-Log Correlation

19. Is there any IP that appears in both auth.log and mail.log?

Answer: 203.0.113.9

20. What was this IP doing? Describe it in 2–3 sentences

Answer:

1. The IP address 203.0.113.9 was performing a coordinated brute-force attack and unauthorized probing across multiple services.
2. In auth.log, it repeatedly attempted to log in to the SSH service using common usernames like root, admin, and pi until it was disconnected for too many authentication failures.
3. In mail.log, the same IP attempted to send unauthorized emails to internal addresses (root@corp.local and postmaster@corp.local), which were rejected as relay attempts.

21. Name one IP in mail.log and dns.log that never appears in auth.log.

Answer: 198.51.100.77

